HB7

Bluetooth Low Energy 5.3 system-on-module

Features

- Bluetooth LE 5.3 with 1 Mbps, 2 Mbps and Coded PHY
- Transmit power up to +8 dBm
- Receiver sensitivity: –97 dBm at 1 Mbps, –103 dBm at 125 kbps (Coded)
- Supply voltage: 1.8 V to 3.6 V (absolute maximum 3.9 V)
- TX current: 6.4 mA at 0 dBm (3 V, DC-DC enabled)
- RX current: 5.1 mA
- Deep sleep current with RTC: 1.2 µA
- 512 kB flash and 128 kB RAM
- Interfaces: SPI, I²C (400 kHz), UART, 12-bit ADC with 8 channels
- Module size 9.5 × 12.5 mm with integrated PCB antenna

General description

The HB7 integrates a BLE 5.3 radio, microcontroller, crystals and antenna in a compact module. It supports over-the-air firmware updates and secure boot.

Applications

Asset trackers, Wearables, Beacons, Smart home sensors.

Ordering information

Order code	Package	Packing	Temperature grade
HB7-MOD-T	Module 9.5×12.5 mm	Tape & reel, 1000	–40 °C to 85 °C

Absolute maximum ratings

Stresses beyond these ratings may cause permanent damage. Functional operation is not implied.

Parameter	Min	Max	Unit
VDD to GND	–0.3	3.9	V
RF input power	—	10	dBm
Storage temperature	–40	125	°C

Recommended operating conditions

Parameter	Min	Typ	Max	Unit
Supply voltage VDD	1.8	3.0	3.6	V
Operating temperature	–40	—	85	°C

Electrical characteristics

Typical values at TA = 25 °C unless otherwise noted.

Parameter	Conditions	Min	Typ	Max	Unit
TX power	Programmable	–20	—	8	dBm
RX sensitivity, 1 Mbps	BER 0.1%	—	–97	—	dBm
RX sensitivity, Coded 125 kbps		—	–103	—	dBm
TX current at 0 dBm	VDD = 3 V, DC-DC	—	6.4	—	mA
RX current	VDD = 3 V, DC-DC	—	5.1	—	mA
Deep sleep current	RTC running	—	1.2	2.0	µA
ADC resolution		—	12	—	bit

Pin configuration and functions

Pin	Name	Type	Description
1	VDD	Power	1.8 V to 3.6 V supply.
2	GND	Ground	Ground.
3	SWDIO	I/O	Serial-wire debug data.
4	SWCLK	Input	Serial-wire debug clock.
5	P0.04/SDA	I/O	GPIO / I²C data.
6	P0.05/SCL	I/O	GPIO / I²C clock.
7	RESET	Input	Active-low reset.

Package and thermal information

Package	Thermal resistance θJA	Dimensions
Module	θJA = 55 °C/W	9.5 × 12.5 mm

Application information

Keep copper and metal away from the antenna end of the module. Use the HM3300 regulator for a low-noise 3.3 V rail when running from a 5 V source.

Revision history

Revision	Date	Change
Rev A	2024-02-20	Initial release
Rev B	2025-01-15	Added Coded PHY sensitivity